

● FINALIZED CARE PLAN

PATIENT NAME

Homo Sapiens (Female,
Neurological & Methylation,
34y)

VISIT DATE

September 30, 2026

PREPARED BY

Pocket Gull AI

BP

108/68

HR

76 bpm

TEMP

98.1°F

SPO2

97%

WEIGHT

138 lbs

HISTORICAL CONDITIONS

Relapsing-Remitting Multiple Sclerosis (RRMS)

History of Optic Neuritis (Left Eye, 2024)

Chronic Fatigue

Mild Depression (reactive)

Vitamin D Deficiency



ACTIVE CARE PLAN

CLINICAL ASSESSMENT

Mara Santos is a 34-year-old female with relapsing-remitting multiple sclerosis (RRMS), diagnosed in 2023 following an episode of left optic neuritis. She is currently on Ocrevus (ocrelizumab) disease-modifying therapy with stable MRI lesion burden. Primary concerns are progressive fatigue, increasing cognitive fog, lower extremity spasticity with gait instability, and reactive depression.

PRIORITY LIST

- Neuroprotection: Maximize myelin preservation and mitochondrial support in the CNS.
- Fatigue Management: Address the multifactorial MS fatigue (central, peripheral, sleep disruption, and deconditioning).
- Spasticity Optimization: Refine Baclofen timing and add non-pharmacological interventions.
- Cognitive Rehabilitation: Structured neuroplasticity exercises to counter cog fog.

- Vitamin D Repletion: Critical immunomodulatory gap — 18 ng/mL is dangerously low for RRMS.

PLAN OF CARE

- Aggressive Vitamin D3 repletion to target 60-80 ng/mL (immune modulation + neuroprotection).
- CoQ10 + Alpha-Lipoic Acid protocol for mitochondrial rescue.
- Structured aquatic therapy program for spasticity + cardiovascular conditioning.
- Cognitive rehabilitation: daily dual-task training (walking + mental arithmetic).
- Reassess NfL and MRI at 6 months to track subclinical disease activity.

GOALS

- Short-term (4 weeks): Fatigue severity scale (FSS) reduction by 20%. Gait confidence improvement.
- Long-term (6 months): Extend relapse-free interval to 18+ months. Serum Vitamin D3 > 60 ng/mL. Stabilize EDSS score.

IMMEDIATE ACTIONS (72 HOURS)

- Begin high-dose Vitamin D3 loading protocol: 10,000 IU/day for 8 weeks, then 5,000 IU maintenance.
- Add CoQ10 200mg BID with meals for mitochondrial support.
- Schedule aquatic therapy evaluation — pool temperature must be < 84°F (Uhthoff's precaution).

MOVEMENT & REHABILITATION PROTOCOL

INTERVENTION	FREQUENCY	DURATION	PRECAUTIONS
Aquatic therapy (cool pool)	3x/week	30 min	Temp < 84°F; monitor for Uhthoff's
Yoga Nidra (guided rest)	Daily	20 min	For fatigue + parasympathetic reset
Dual-task gait training	5x/week	15 min	Walking + cognitive tasks simultaneously
Progressive resistance (seated)	2x/week	20 min	Target proximal muscle groups

SPASTICITY MANAGEMENT

- Baclofen timing optimization: shift from TID to QID with lower individual doses (7.5mg x4) to smooth coverage.
- Add evening stretching routine (hamstrings, hip flexors, calves) — 15 min before bed.
- Consider trial of CBD oil (15mg sublingual) for breakthrough spasticity — discuss with neurologist.

HEAT SENSITIVITY PROTOCOL

- Cooling vest for any outdoor activity > 75°F.
- Pre-cooling before exercise (cold towel on neck, cold water).
- Avoid hot baths/showers; use lukewarm water only.

BIOCHEMICAL ASSESSMENT

Mara's nutritional profile reveals significant micronutrient depletion consistent with active autoimmune neuroinflammation. Vitamin D deficiency is the most critical gap — studies show RRMS patients with serum D3 > 50 ng/mL have 50-70% fewer relapses. Glutathione depletion signals oxidative stress in the CNS.

NUTRITION TARGETS

- Vitamin D3: Aggressive repletion to 60-80 ng/mL — immunomodulatory priority.
- Anti-inflammatory Diet: Mediterranean-style with emphasis on omega-3 fatty acids.
- Gut-Brain Axis: Microbiome support for immune regulation.

NUTRITIONAL INTERVENTIONS

NUTRIENT/COMPOUND	THERAPEUTIC DOSE	DELIVERY	TARGETED PATHWAY
Vitamin D3 + K2	10,000 IU / 200mcg MK-7	Oral with fat	T-reg upregulation / Neuroprotection
Omega-3 EPA/DHA	3000mg (high EPA)	Oral with meals	Neuroinflammation / Myelin integrity
N-Acetyl Cysteine (NAC)	600mg BID	Oral	Glutathione precursor / Antioxidant
Probiotics (multi-strain)	50B CFU	Oral AM	Gut-brain axis / Immune regulation

DIETARY PATTERN

- Adopt: Modified Mediterranean / Wahls Protocol Level 1 — 9 cups vegetables daily (3 cups greens, 3 cups sulfur-rich, 3 cups deeply colored)
- Eliminate: Gluten, dairy, refined sugar (30-day elimination trial to assess neuroinflammatory response)
- Emphasize: Wild-caught fatty fish 3x/week, bone broth daily, fermented vegetables

BIOCHEMICAL & BIOMARKER MATRIX

Mara's orthomolecular profile is characteristic of active CNS autoimmune disease: severe Vitamin D deficiency driving immune dysregulation, depleted glutathione indicating oxidative stress on oligodendrocytes, low CoQ10 impairing mitochondrial respiration in demyelinated axons, and elevated homocysteine suggesting methylation pathway strain.

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DETECTED DEFICIENCIES & MS RELEVANCE

- Vitamin D3 (Deficient at 18 ng/mL): The most important modifiable risk factor in RRMS. Low D3 is associated with higher relapse rates, faster disability accumulation, and increased MRI lesion burden. Target: 60-80 ng/mL.
- Glutathione (Low at 1.1 µmol/g Hb): Oligodendrocytes (the cells that make myelin) are exquisitely sensitive to oxidative stress. Depleted GSH accelerates demyelination.
- CoQ10 (Low at 0.38 µg/mL): Demyelinated axons require dramatically more energy to conduct signals. Low CoQ10 = mitochondrial failure = axonal degeneration.
- Homocysteine (Elevated at 14.1 µmol/L): Neurotoxic at elevated levels; impairs myelin synthesis and increases BBB permeability.

PRECISION NUTRIENT PROTOCOL

INTERVENTION/MOLECULE	THERAPEUTIC DOSE	DELIVERY	TARGETED PATHWAY
Vitamin D3 + K2	10,000 IU / 200mcg MK-7	Oral with fat	T-reg upregulation / Neuroprotection
N-Acetyl Cysteine (NAC)	600mg BID	Oral	Glutathione precursor / Oligodendrocyte shield
CoQ10 (Ubiquinol)	200mg BID	Oral with fat	Mitochondrial Complex I-III rescue
Methylcobalamin (B12)	1000mcg sublingual	Daily	Myelin synthesis / Homocysteine clearance

INTERVENTION/MOLECULE	THERAPEUTIC DOSE	DELIVERY	TARGETED PATHWAY
L-Methylfolate (5-MTHF)	800mcg	Oral	Methylation cycle / Homocysteine clearance
Alpha-Lipoic Acid (R-ALA)	600mg	Oral	BBB-crossing antioxidant / Neuroprotection
Magnesium Glycinate	400mg	Oral, nightly	NMDA modulation / Spasticity / Sleep

CAUTIONS & INTERACTIONS

- Vitamin D3 + Ocrevus: Monitor calcium levels quarterly; immunosuppression + high-dose D3 requires careful calcium homeostasis surveillance.
- NAC + Escitalopram: NAC may modulate glutamatergic neurotransmission; generally synergistic with SSRIs but monitor for mood changes.
- Alpha-Lipoic Acid + Blood Sugar: R-ALA can lower blood glucose; monitor if Mara ever starts corticosteroids for relapse treatment.
- B12 Supplementation: Use methylcobalamin form specifically — cyanocobalamin requires hepatic conversion that may be impaired in MS.

IMMEDIATE MONITORING (0-30 DAYS)

1. Baseline Fatigue Severity Scale (FSS) and Modified Fatigue Impact Scale (MFIS) scores.
2. Recheck serum 25(OH)D at 8 weeks — target > 60 ng/mL.
3. Begin daily fatigue diary in Pocket Gull (rate 1-10, morning + afternoon).
4. Monitor Baclofen dose adjustment — watch for increased drowsiness.

ONGOING SURVEILLANCE

PARAMETER	TARGET	FREQUENCY	ESCALATION TRIGGER
Serum Vitamin D3	60-80 ng/mL	8 weeks, then quarterly	< 30 ng/mL or > 100 ng/mL
Neurofilament Light (NfL)	< 10 pg/mL	Every 6 months	Rising trend = subclinical activity
EDSS Score	Stable or improving	Every 6 months	Any 0.5-point increase
MRI Brain + C-spine	No new T2 lesions	Annual (or if symptoms)	New enhancing lesions
Fatigue Severity Scale	< 4.0	Monthly	> 5.5 sustained
25-ft Walk Test	Stable or improving	Every 3 months	> 20% decline

RED FLAGS — SEEK IMMEDIATE CARE

⚠️ New or worsening visual loss, sudden limb weakness, bladder retention, or severe vertigo lasting > 24 hours may indicate a relapse. Contact your MS neurologist immediately — do NOT wait for a scheduled visit.

LONG-TERM TRAJECTORY

With Ocrevus maintaining immune suppression, aggressive Vitamin D repletion, mitochondrial support, and structured rehabilitation, Mara's trajectory should show stabilization of EDSS and extension of the relapse-free interval. The elevated NfL (18.2 pg/mL) warrants close 6-month surveillance.

UNDERSTANDING MULTIPLE SCLEROSIS

Your immune system is mistakenly attacking the myelin coating on your nerve fibers — think of it like the insulation on electrical wires getting damaged. When the insulation is thin or missing, the signals slow down or short-circuit. That's why you experience the fatigue, the fog, the spasticity, and the tingling.

WHAT YOUR CARE PLAN ADDRESSES

- **The Fatigue:** MS fatigue isn't just 'being tired.' Your brain is working harder to route signals around damaged areas — like a GPS rerouting around road closures. The CoQ10 and mitochondrial support help your nerve cells produce more energy for this extra work.
- **The Spasticity:** Your muscles are getting garbled signals, causing them to tighten when they shouldn't. Baclofen helps quiet these signals, and aquatic therapy gives your body a low-gravity environment to move freely.
- **The Cognitive Fog:** Your brain can rewire itself (neuroplasticity). The dual-task exercises are training your brain to build new pathways around the damaged ones.
- **Vitamin D:** This is not just a vitamin for you — it's a critical immune system regulator. Keeping your levels high has been shown to significantly reduce MS relapses.

YOUR ACTION PLAN

- **Today:** Start Vitamin D3 with your next meal. This is the single most impactful thing you can do right now.
- **This Week:** Begin the evening stretching routine for spasticity. Try the Yoga Nidra recording for fatigue.
- **This Month:** Schedule aquatic therapy evaluation. Start the fatigue diary in the app.

IMPORTANT SAFETY NOTES

-  Never skip your Ocrevus infusions — they are the foundation protecting your brain from new attacks. Everything else we're doing is designed to work alongside your DMT, not replace it.
-  Heat is your enemy. Always have a cooling strategy before exercise or hot days. Even a warm shower can temporarily worsen your symptoms (Uhthoff's phenomenon) — this is not a relapse, it's temporary.

PERINATAL HEALTH ASSESSMENT

Homo Sapiens (Female, Neurological & Methylation, 34y) is a 34-year-old female evaluated for perinatal functional optimization and maternal autonomic balance. Maternal blood pressure is 108/68 with stable SpO2 at 97%.

PERINATAL SAFETY THRESHOLDS

Metric	Target Range	Frequency	Action Threshold
Blood Pressure	< 130/80 mmHg	Daily	> 140/90 mmHg (Preeclampsia Rule-out)
Fetal Movement	≥ 10 kicks / 2 hrs	Daily	< 10 kicks in 2 hours
Postpartum Mood	EPDS Score < 9	Bi-weekly	EPDS ≥ 10

INTEGRATED CARE PROTOCOL

- Western Support: Pre-eclampsia monitoring, prenatal iron/folate supplementation.
- Eastern TCM: Ren Mai (Conception Vessel) warming & Shen calming.
- Ayurvedic Garbhini: Sattvic diet, warm sesame Abhyanga, Nadi Shodhana breathing.

ECOLOGICAL CELLULAR & GARDEN METAPHORS

Your cellular energy system operates like a living garden ecosystem. Nutrients and hydration are your fertile soil; circadian sleep and morning sunlight provide vital solar energy.

ANCESTRAL DIETARY HERITAGE & CULTURAL SWAPS

Cultural Heritage	Traditional Swap	Bio-Individual Benefit	Local Sourcing Tip
African Heritage	Collard greens & okra broth	High folate, prebiotic fiber	Local farmers market
Latino/Hispanic	Black beans & avocado oil	Stable glycemic index, oleic acid	Standard grocery produce
East/South Asian	Miso & ginger jujube tea	Probiotic enzymes, digestion	Asian specialty market

DAILY MICRO-HABITS (GROW-THYSELF QUESTS)

- 🌅 Morning Circadian Sun Reset: 15 min morning light.
- 🫁 Vagal Tone Breathing: 6.0 bpm resonant breathing.
- 🍵 Ancestral Mineralization: Warm herbal mineral infusion.

ACTUARIAL RISK & BIOLOGICAL AGE SYNTHESIS

Quantitative Gompertz-Makeham hazard modeling indicates a biological age delta of -4.5 years relative to chronological age (34), projecting +7.2 Quality-Adjusted Life Years (QALY) gain.

CDC 4-DRIVER HAZARD REDUCTION MATRIX

MORTALITY VECTOR	BASELINE RISK	MULTI-PARADIGM STRATEGY	PROJECTED HAZARD RATIO (HR)
Cardiovascular	Moderate	HRV 6.0 bpm Vagal Tone + CoQ10	0.62 (38% Risk Reduction)
Metabolic	Low-Moderate	Sirtuin Activators + Low Glycemic	0.55 (45% Risk Reduction)
Neurodegenerative	Low	TCM Shen Harmonization + Neti/Nasya	0.68 (32% Risk Reduction)
Oncological	Low	Autophagic Rasayana & Anti-inflammatory	0.74 (26% Risk Reduction)

MULTI-PARADIGM CELL LONGEVITY PROTOCOL

- Western Epigenetic Support: Sirtuin activation (NAD+/NMN), AMPK activation, HRV vagal tone.
- Eastern TCM Essence: Zang-Fu meridian balancing, sleep architecture, Jing preservation.
- Ayurvedic Rasayana: Ama detoxification, Agni metabolic fire, adaptogenic rejuvenation.

PRE-CONCEPTION PARENTAL EPIGENETIC MATRIX

OPTIMIZATION VECTOR	TARGETED BIOMARKER	PARENTAL PROTOCOL	FETAL HEALTHSPAN IMPACT
Folate Methylation	MTHFR C677T / Homocysteine	L-5-MTHF (800 mcg) + B12	Neural tube & DNA fidelity
Maternal Mitochondria	CoQ10 Ubiquinol / Fasting Insulin	Ubiquinol (300 mg) + Alpha-Lipoic	High oocyte cleavage ATP
Paternal Spermatogenesis	Sperm DNA DFI / Zinc	Zinc (30 mg) + Selenium + Carnitine	Lower germline mutational load
Environmental Detox	Heavy metals & Endocrine Disruptors	Filtered water & organic whole foods	Reduced epigenetic transgenerational risk

SENIOR & GERIATRIC CARETAKING STRATEGY

- Autonomic Vagal Tone: HRV SDNN 6.0 bpm resonant breathing sessions to reduce agitation and improve sleep.
- TCM Kidney Jing & Shen: Warm bone broths, Rehmannia, and Ginkgo Biloba to nourish brain marrow (*Sui Hai*).
- Ayurvedic Vata Pacification: Warm sesame Abhyanga massage & Bacopa Monnieri (Brahmi) for joint/cognitive support.

VOLUNTARY ACOG/ACMG CARRIER SCREENING GUIDANCE

Recommends voluntary pre-conception carrier screening for autosomal recessive traits. Pre-conception optimization ensures offspring inherit optimal epigenetic programming while preserving human dignity.

 CARE PLAN INSTRUCTIONS

- ☐ [Goal]: Optimize highly effective CFTR modulator therapy (elexacaftor/tezacaftor/ivacaftor), airway clearance vibrating vest, and high-calorie pancreatic enzyme dosing.

 OXIDATIVE STRESS MARKERS

- hsCRP: 3.2 mg/L (Elevated)
- Homocysteine: 14.1 µmol/L (Elevated)
- Neurofilament Light Chain (NfL): 18.2 pg/mL (Elevated)

 ANTIOXIDANT SOURCES

- Glutathione (GSH): 1.1 µmol/g Hb (Low)
- CoQ10: 0.38 µg/mL (Low)
- Alpha-Lipoic Acid: Not tested

 MEDICATIONS

- Ocrevus (Ocrelizumab): 600mg IV q6mo
- Baclofen: 10mg TID (spasticity)
- Modafinil: 100mg AM (fatigue)
- Escitalopram: 10mg Daily

CLINICIAN SIGNATURE

DATE & TIME

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